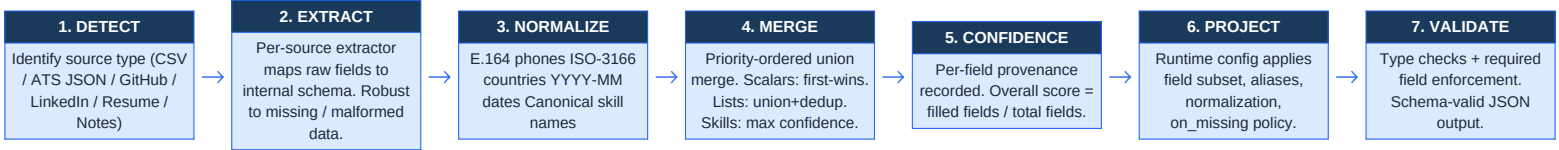


Multi-Source Candidate Data Transformer

Eightfold Engineering Intern · Jul-Dec 2026 · Technical Design Document

PIPELINE ARCHITECTURE



CANONICAL OUTPUT SCHEMA

<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Field' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='H elvetica-Bold', fontSize=6.8, greek=0, italic=0, link=[], rise=0, text='Field', textC olor=Color(.121569,.1607 84,.215686,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Type' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helve tica-Bold', fontSize=6.8, greek=0, italic=0, link=[], rise=0, text='Type', textC olor=Color(.121569,.1607 84,.215686,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Normalization' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=6.8, greek=0, italic=0, link=[], rise=0, text='Normalization', textC olor=Color(.121569,.160784,.215686,1), us_lines=[])]) 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>
candidate_id	string	SHA-256 of primary email
full_name	string	—
emails	string[]	lowercase, deduped
phones	string[]	E.164 (e.g. +919876543210)
location	{city,region,country}	ISO-3166 alpha-2 country
links	{linkedin,github,...}	URL as-is
headline	string null	—
years_experience	number null	parsed from text
skills	[[name,conf,sources[]]]	Canonical skill names
experience	[[company,title,start,end]]	dates: YYYY-MM
education	[[{institution,degree,field,end_year}]]	—
provenance	[[field,source,method]]	per-field traceability
overall_confidence	number	0.0–1.0

MERGE / CONFLICT POLICY

- Source priority (high → low)**
LinkedIn > ATS JSON > CSV > GitHub > Resume > Recruiter Notes
- Scalars (name, headline, years_exp):**
- First non-null value from highest-priority source wins
- Lists (emails, phones):**
- Union across all sources, deduped
 - Emails: lowercased before dedup
 - Phones: E.164 normalized before dedup
- Skills:**
- Union by canonical name
 - Confidence = max across sources
 - sources[] = union of all reporting sources
- Experience / Education:**
- Dedup by (company, title) or (institution, degree)
 - First-seen entry wins; no field merging within entries

EDGE CASES HANDLED

- Missing source:** skipped silently, error logged in `_extraction_errors`
- Malformed JSON/CSV:** extractor returns `{}` — pipeline continues
- Conflicting names:** priority order resolves, provenance recorded
- Phone without country code:** assumes +91 (India) for 10-digit
- Unknown skill:** title-cased as-is; not invented
- Duplicate experience:** deduped by (company_lower, title_lower)
- required field missing + on_missing=error:** raises ValueError
- Empty source list:** returns structured error dict, not crash

DELIBERATE DESCOPIES

- No real LinkedIn scraping (requires auth); accepts pre-parsed dict
- GitHub language confidence is heuristic (repo count), not skill depth
- Resume NLP is regex/keyword — no ML model in this phase
- No dedup across experience entries from same source (first-seen wins)

RUNTIME CONFIG (PROJECTION LAYER)

A config JSON reshapes output without touching the engine:

fields[]: path (e.g. "emails[0]", "skills[].name"), from (output alias), type, required, normalize ("E164" | "canonical")

- include_confidence: true/false
- include_provenance: true/false
- on_missing: "null" | "omit" | "error"

Example config:

```
{  
  "fields": [  
    {  
      "path": "emails[0]",  
      "from": "primary_email",  
      "type": "string",  
      "required": true,  
      {  
        "path": "skills[].name",  
        "normalize": "canonical"  
      }  
    },  
    {  
      "path": "skills[].name",  
      "normalize": "canonical"  
    }  
  ],  
  "include_confidence": true,  
  "on_missing": "null"  
}
```

Implementation: Python 3.10+ · stdlib only (no third-party deps for core) · CLI: `python src/cli.py --csv ... --ats ... --github ... --config ...` · 55 unit tests · 100% pass rate